

MARK SCHEME for the May/June 2006 question paper

9702 PHYSICS

9702/06

Paper 6

Maximum raw mark 40

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. It shows the basis on which Examiners were initially instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began. Any substantial changes to the mark scheme that arose from these discussions will be recorded in the published *Report on the Examination*.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the *Report on the Examination*.

The minimum marks in these components needed for various grades were previously published with these mark schemes, but are now instead included in the *Report on the Examination* for this session.

- CIE will not enter into discussion or correspondence in connection with these mark schemes.

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Page 1	Mark Scheme	Syllabus	Paper
	GCE A – May/June 2006	9702	06

Option A - Astrophysics and Cosmology

- 1 Planet: almost circular orbits
all in nearly the same plane

Comet: highly elliptical orbits
in many different planes
- 2 (a) (mean) density
of matter in the Universe
- (b) (i) symmetrical curve below given line
touching given line at 'present time'
- (ii) H_0 not known with any certainty
mass of matter in the Universe not known
extent of Universe unknown
(allow 1 of the last 2 marks for ρ_0 not known)
- 3 1 light-year = 0.306 pc (allow 0.3 pc)
 1.3×10^{10} light-years = 3.98×10^3 Mpc
 $v = H_0 d$
speed = $60 \times 3.98 \times 10^3 = 2.39 \times 10^5 \text{ km s}^{-1}$
ratio = $(2.39 \times 10^5 \times 10^3) / (3.0 \times 10^8)$
= 0.8
- 4 e.g. vast expense
money could be spent on humanitarian aid
- observations possible that cannot be made on Earth
since atmosphere limits observations
- technological/scientific developments on Earth
greater understanding of Universe
leads to 'spin off' benefits for individuals
- Any sensible comments, 1 each to max 5

Option F - The Physics of Fluids

- 5 (a) conservation of volume/mass/density or incompressible
- (b) conservation of energy
- 6 (a) air near jet is moving at speed
higher speed air has a lower pressure
(because) air is dragged along by water jet
air (outside pump) is not moving
- OR water in jet is moving at speed
OR high-speed water has lower pressure

OR air is drawn into water jet

OR loss of air reduces pressure
- (b) (i) air/water in pump has a higher speed
so greater pressure difference

Page 2	Mark Scheme	Syllabus	Paper
	GCE A – May/June 2006	9702	06

- (ii) no change in speed of air OR reference to greater ρ in Bernoulli eqn **M1**
so no change in pressure OR greater pressure difference **A1** [2]
difference

(allow any logical argument based on liquid causing more/less drag on air)

- 7 (a) eddy currents have kinetic energy OR cause extra drag **M1**
eddy currents caused by movement of the car OR energy required to overcome drag **A1**
extra energy (of eddy currents) is derived from car's fuel **A1** [3]
- (b) (i) power = force $\times$ speed **B1**
so power = $\frac{1}{2}C_D A \rho v^2 \times v$ and A and ρ are constants **B1** [2]
- (ii) $84 \times 10^3 = \frac{1}{2} \times 0.34 \times 1.8 \times 1.1 \times v_{\max}^3$ **C1**
 $v_{\max} = 63 \text{ m s}^{-1}$ **A1** [2]
- (iii) $P = \frac{1}{2} \times 0.34 \times 1.8 \times 1.1 \times (63 + 9)^3$ **C1**
 $P = 126 \text{ kW}$ **C1**
ratio = $126 / 84 = 1.5$ **A1** [3]

Option M - Medical Physics

- 8 (a) alternating voltage **B1**
applied across (piezo-electric) crystal **B1**
causes crystal to vibrate **B1**
crystal dimensions such as to give resonance (in US range) **B1** [4]
- (b) wavelength at 1 MHz is shorter **B1**
so greater detail is possible **B1** [2]
- 9 e.g. used as a scalpel (1)
further detail: causes (explosive) vaporisation of intracellular water (1)
CO₂ laser (1)
IR radiation strongly absorbed by water (1)
laser beam focused to give high power density (1)
no/very little bleeding (1)
accurate guidance (1)
- e.g. repair of retina (1)
further detail: focused laser beam onto retina (1)
melts tissue and forms a weld (1)
(pulsed) ruby or argon laser (1)
- any two examples: named (1) plus further detail (2)* **B6** [6]
- (allow up to two marks for each diagnostic technique)*
- 10 (a) minimum intensity (of sound) detected **M1**
where intensity = (sound) power per unit area at a stated frequency **A1**
value is $1 \times 10^{-12} \text{ W m}^{-2}$ **B1**
at 3 kHz (*allow 2 kHz $\rightarrow$ 3 kHz*) **B1** [4]

Page 3	Mark Scheme	Syllabus	Paper
	GCE A – May/June 2006	9702	06

- (b) (i) intensity = $(0.14 \times 10^{-6})/(54 \times 10^{-6}) = 2.6 \times 10^{-3} \text{ W m}^{-2}$ C1
 $I/L = 10 \lg (2.6 \times 10^{-3})/(1 \times 10^{-12})$ C1
= 94 dB A1 [3]

- (ii) comment e.g. would be perceived as being loud
could cause tinnitus over a short period of time
could cause deafness over a long period of time
higher level than is acceptable in the workplace

any appropriate comment, 1 mark

B1 [1]

Option P - Environmental Physics

- 11 (a) at times of low usage of electrical power B1
water pumped from low-level to high-level reservoir B1
at times of high/sudden demand for electrical power B1
water released to pass through turbines B1 [4]

- (b) electrical energy generated = $78 \times 10^6 \times 4.0 \times 3600 = 1.12 \times 10^{12} \text{ J}$ C1
energy to be stored = $(1.12 \times 10^{12})/0.75 = 1.5 \times 10^{12} \text{ J}$ C1
 $1.5 \times 10^{12} = \rho Vgh$ C1
 $= 1.0 \times 10^3 \times V \times 9.8 \times 95$
 $V = 1.6 \times 10^6 \text{ m}^3$ A1 [4]

- 12 (a) law: it is impossible to convert all of a given amount of thermal energy into work B1
(that is) $W < Q_H$ B1
 $(Q_H - W)$ is energy rejected at temperature T_L B1 [3]

- (b) $W/Q_H = 1 - T_L/T_H$ B1 [1]

- (c) efficiency = $1 - 313/393$ C1
= 0.20 A1 [2]

- 13 (a) (i) e.g. industry setting up
people preparing to go to work
starting to cook breakfast

(allow any two sensible suggestions, 1 each)

B2 [2]

- (ii) e.g. change in temperature with use of heaters/air conditioning
holiday or workday with more power used by industry when not on holiday

(allow any two sensible suggestions, 1 each)

B2 [2]

Page 4	Mark Scheme	Syllabus	Paper
	GCE A – May/June 2006	9702	06

- (b) (i) sudden increase in demand (as appliances are used) B1
- (ii) increased demand in the afternoon B1 [2]
- (allow any two sensible suggestions in (i) and (ii))

Option T - Telecommunications

- 14 (a) (instantaneous) displacement of information signal determines the frequency of the carrier wave M1 A1 [2]
- (b) (i) 12 V B1 [1]
- (ii) 650 kHz B1 [1]
- (iii) 550 kHz B1 [1]
- (iv) 3000 B1 [1]
- 15 (a) analogue-to-digital converter (*do not allow ADC*) B1 [1]
- (b) controls the time at which samples are taken B1 [1]
- (c) enables higher frequency components in signal to be 'detected' B1 [1]
- 16 (a) electromagnetic shielding for the inner conductor the braid is earthed B1 B1 [2]
- (b) increased bandwidth means more information can be carried so more calls can be transmitted simultaneously fewer links are required B1 B1 B1 [3]
- 17 (a) e.g. cross-talk/cross-linking interference/picking up atmospherics/picking up man-made radiation white noise associated with vibrating atoms (any two, 1 each) B2 [2]
- (b) (i) number of dB = $10 \lg (P_2/P_1)$
 $35 = 10 \lg (P/\{7.6 \times 10^{-6}\})$
 $P = 0.024 \text{ W}$ C1 A1 [2]
- (ii) number of dB = $10 \lg (2.6/0.024) = 20.3$
length = $20.3/5.8 = 3.5 \text{ km}$ C1 A1 [2]